

Choose any 12 problems.

1. Determine the following:

- a. Write the following set in tabular form: $A = \{x: x \text{ is a letter in "do not repeat"}\}$.
- b. Write the following set in tabular form: $B = \{x: x = 2n \text{ for } n \in \mathbb{Z}\}$.
- c. Write the set $C = \{\dots, -3, 0, 3, 6, \dots\}$ in set-builder form.

2. Give the definition of the following:

- d. A *function*.
- a. A *1-1 function*.
- b. A *transitive relation*.

3. Complete the following:

- a. Find all the subsets of $A = \{a, b, c\}$.
- b. If $|A| = 4$ how many subsets does A have?

4. Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$.

- a. Find $A \cup B$.
- b. Find $A \cap B$.
- c. Find $A - B$.
- d. Find $A \times B$.

5. Let $A \subseteq B$ and $B \subseteq C$. Prove that $A \subseteq C$.

6. Prove that if $A \subseteq B$ then $A \cup B = B$.

7. Prove that $(A \cap B)^c \subseteq A^c \cup B^c$.

8. Let $f : A \rightarrow B$ be a function from set A to set B .

a. If $|A| = 20$ and $|B| = 21$ can f be one-to-one?

b. If $|A| = 20$ and $|B| = 21$ can f be onto?

c. If f is a constant function and $|B| > 1$, can it be 1-1? (f is *constant function* if $f(a) = b$ for all $a \in A$ and some $b \in B$.)

9. Complete the following:

d. Give an example of a function which is not one-to-one.

e. Give an example of a function which is not onto.

10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^3$.

a. Prove that f is one-to-one.

b. Prove that f is onto.

11. Let $\mathcal{R} = \{(a, b), (b, a), (b, c), (a, a), (b, b), (c, c)\}$ be a relation on the set $E = \{a, b, c\}$.

c. Is $\mathcal{R}$ reflexive? Explain.

d. Is $\mathcal{R}$ symmetric? Explain.

12. Define the relation $\mathcal{R}$ on the set of even integers as $x\mathcal{R}y \iff 2|x \cdot y$ (i.e., $x \cdot y$ is an even integer). Show that $\mathcal{R}$ is an equivalence relation.

13. Complete the following:

a. Is $16 \equiv 4 \pmod{12}$?

b. Is $5 \equiv 4 \pmod{12}$?

c. Find an integer x such that $0 \leq x < 4$ and $x \equiv 15 \pmod{4}$.